On-Device Speech Scoreboard

An Android app that writes down what was said in a conversation and who said it, running entirely on the phone with no network connection. These are its accuracy results in English and German, on two tablets.

ElevenLabs voices | Lenovo TB336FU · MediaTek MT8755 | Xiaomi 25097RP43I · Snapdragon 8s Gen 4 | 2026-08-25 to 08-27

HOW A RECORDING BECOMES A LABELLED TRANSCRIPT

Audio is captured at 16 kHz mono. Five stages run over it, and the results exercise all of them at once.

1. A segmentation model cuts the audio into turns, meaning stretches where one person is talking. It treats silence as the cue for where a turn ends.
2. An embedding model converts each turn into a voiceprint: a vector describing the voice itself rather than the words.
3. The voiceprints are clustered. With no speaker count given, clustering cuts at a 0.5 similarity threshold; small fragments are then folded into larger clusters.
4. Each cluster is compared against the voiceprints of people who enrolled and takes the closest name. Clusters with no close match stay unnamed.
5. A recogniser transcribes the audio, and its word timings are matched to the turn ranges, so every word ends up underneath a speaker.

Enrolment means recording a few seconds of someone speaking so the app has a voiceprint to compare against later.

Cutting silence is optional and changes what stage 1 receives: the silent parts are removed first and only speech reaches attribution, saving memory and time on a recording with long pauses. The recogniser still gets the whole recording, because a word the detector missed should still be transcribed. The "cut" column reports how much was removed.

WHAT WAS USED

Each model is downloaded once and runs locally through sherpa-onnx, 683 MB together. Parakeet is used because attribution needs per-word timings for stage 5, and it is one of only two recognisers available here that report them.

STAGE	MODEL	SIZE	RUNS ON
Turn segmentation	pyannote segmentation 3.0 sherpa-onnx-pyannote-segmentation-3-0	5.7 MB	CPU
Voiceprints	3D-Speaker ERes2Net-base 3dspeaker_speech_eres2net_base_sv_zh-cn_3dspeaker_16k	37.8 MB	CPU
Transcription	NVIDIA Parakeet TDT 0.6B v3, int8 sherpa-onnx-nemo-parakeet-tdt-0.6b-v3-int8	639 MB	CPU

Everything runs on the CPU. The speech library ships prebuilt with ONNX Runtime compiled in, and the copy shipped for 64-bit ARM contains 172 references to the CPU execution provider and none at all to XNNPACK, NNAPI or QNN.

THE VOICES

The audio was synthesised by ElevenLabs, whose voices are close enough to real speech that telling two apart is a fair test. Three appear: Sarah, a woman, and Daniel and George, both men. Each turn was synthesised separately and the pieces joined, so the correct answer is known to the exact audio sample: who spoke, from when, to when.

The pairing matters more than the individual voices. A man and a woman is the easy case. Two men is the hard one, for a measurable reason: the two male voiceprints sit at **0.734 cosine similarity**, against **0.91** between two separate recordings of the same person. That gap is the ceiling every later stage inherits, because once a cluster contains both voices only one name can attach to it and no later stage recovers the other.

Coverage
How much of the expected text the transcript reached. Read before the error rate: below 90% the transcript stopped early and its error rate means nothing.

WER
Word error rate. Share of words the recogniser got wrong, numerals spelled out so "14001" and "fourteen thousand and one" agree. Lower is better.

Speaker, by word
Of words both transcripts agree were said, the share placed under the right speaker. The figure the app displays.

Frame
Share of 10 ms slices of speech attributed correctly. Weights a speaker by how long they talked.

Turn
Share of turns whose majority speaker is right. Every turn counts once however short, so this is the harshest measure and the closest to how a reader experiences the transcript.

RESULTS IN ENGLISH AND GERMAN

One 36-turn auditor and employee conversation. The German file is the same conversation, 24 turns, spoken by the same two voices.

FILE	VOICES	LG	DEVICE	CUT	AUDIO	TIME TAKEN	COVERAGE	WER	SPEAKER	FRAME	TURN
identifiable_fm one man, one woman, the easy case	11L Sarah f, Daniel m	EN	Lenovo	not logged	4m 44s	2m 44s (1.73×)	99.1%	2.5%	98.3%	96.2%	**29/36**
eleven_two_voice two men, the hard case	11L George m, Daniel m	EN	Lenovo	not logged	4m 49s	2m 50s (1.70×)	99.1%	2.9%	98.6%	96.0%	27/36
eleven_two_voice two men, the hard case	11L George m, Daniel m	EN	Lenovo	not logged	4m 49s	6m 22s (0.76×)	99.1%	2.9%	**99.0%**	**96.8%**	**29/36**
eleven_two_voice two men, the hard case	11L George m, Daniel m	EN	Xiaomi	0.1%	4m 49s	1m 00s (4.79×)	99.1%	2.9%	**99.0%**	**96.8%**	**29/36**
eleven_ota same file played aloud and re-recorded	11L George m, Daniel m	EN	Lenovo	none	5m 11s	2m 46s (1.87×)	99.3%	2.5%	98.5%	95.9%	26/36
eleven_ota same file played aloud and re-recorded	11L George m, Daniel m	EN	Xiaomi	**7.4%**	5m 11s	1m 28s (3.53×)	99.3%	2.5%	98.3%	96.0%	**29/36**
eleven_de the same conversation in German	11L George m, Daniel m	DE	Lenovo	0.4%	3m 05s	2m 07s (1.46×)	98.2%	2.5%	98.2%	95.3%	16/24
eleven_de the same conversation in German	11L George m, Daniel m	DE	Xiaomi	0.4%	3m 05s	37s (4.96×)	98.2%	2.5%	98.2%	95.3%	16/24
eleven_two_voice probe: every pause cut out by hand	11L George m, Daniel m	EN	Lenovo	3.3%	4m 39s	2m 42s (1.73×)	99.0%	2.5%	98.8%	95.5%	27/36
eleven_meeting_ota 40.5% dead air, played aloud and re-recorded	11L George m, Daniel m	EN	Lenovo	**36.7%**	7m 49s	3m 38s (2.15×)	98.4%	2.8%	95.8%	93.1%	26/36
MEAN, 10 RUNS	THREE VOICES	EN+DE	BOTH	N/A	**48m 10s**	**26m 15s** (1.84×)	**98.9%**	**2.65%**	**98.3%**	**95.7%**	**254/336 · 75.6%**

Both tablets gave the same numbers
English and German matched across devices down to the frame count: 27,009 of 27,910 speech frames on both, 27 blocks on both, the same 29 turns of 36. German matched at 17,016 of 17,846 frames with identical clustering, four clusters folding to two. Attribution is deterministic given the same audio and models. The two English rows reading 98.6% and 99.0% are two enrolments, not two devices; both tablets reproduced both values exactly.

Reading the mean
The mean is unweighted across ten runs differing in device, language and purpose, so it summarises this table rather than stating a headline figure. Turn accuracy is pooled, 254 correct of 336.

Sources: docs/diarization-benchmark.md, docs/data/README.md. Scored with diarscore.py and wer.py. Re-scored from the devices' own databases, 2026-08-27.